Customer Shopping Behavior Analysis

Uncovering insights from 3,900 purchases to guide strategic business decisions

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Dataset Overview

3,900

Purchase Records

Transactional data analyzed

18

Data Columns

Customer and purchase details

37

Missing Values

Review ratings imputed

Key Features

- Customer demographics
- Purchase details
- Shopping behavior

Analysis Tools

- Python for EDA
- SQL for queries
- Power BI dashboard

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Data Preparation Process

Data Loading

Imported dataset using pandas

Initial Exploration

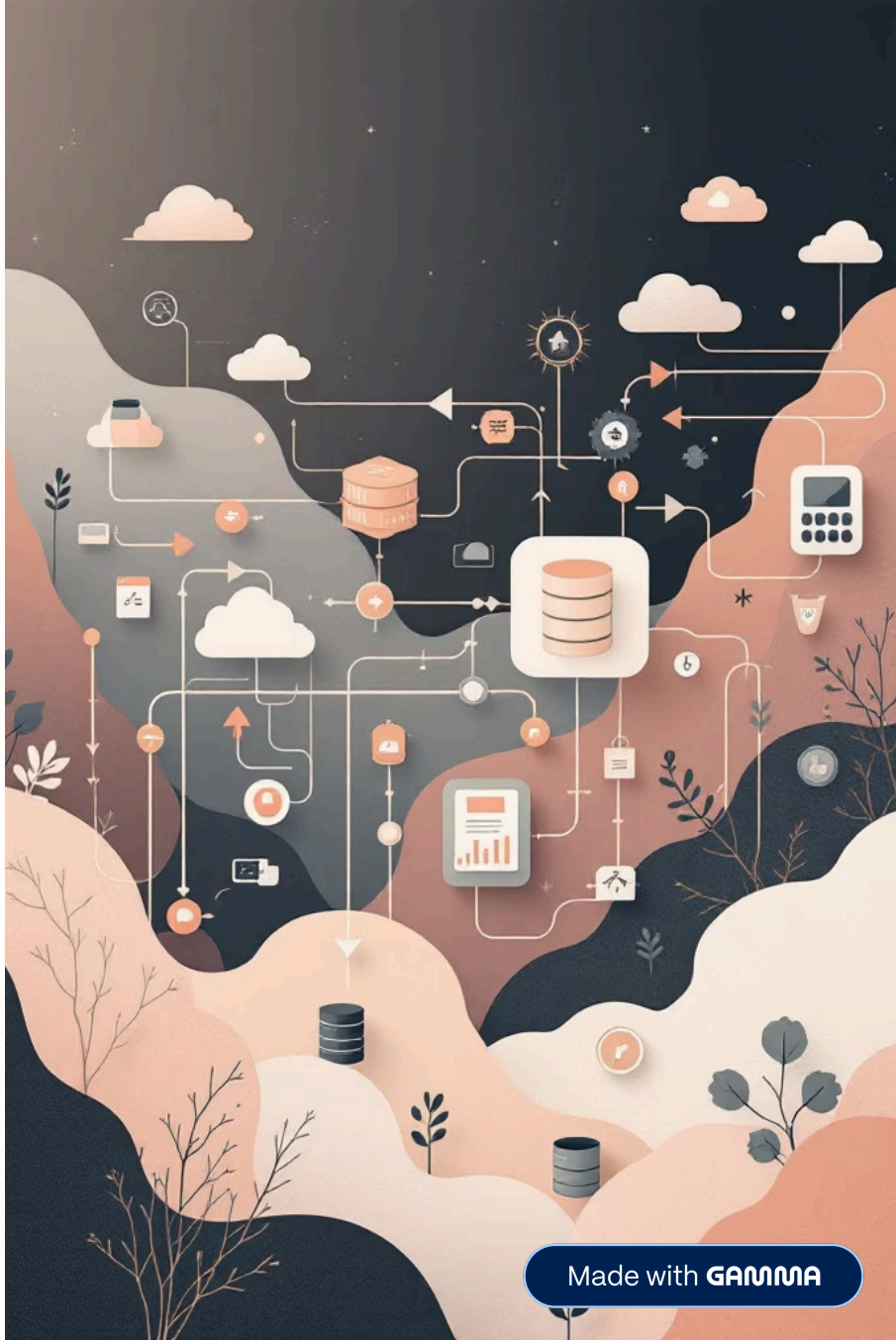
Checked structure and summary statistics

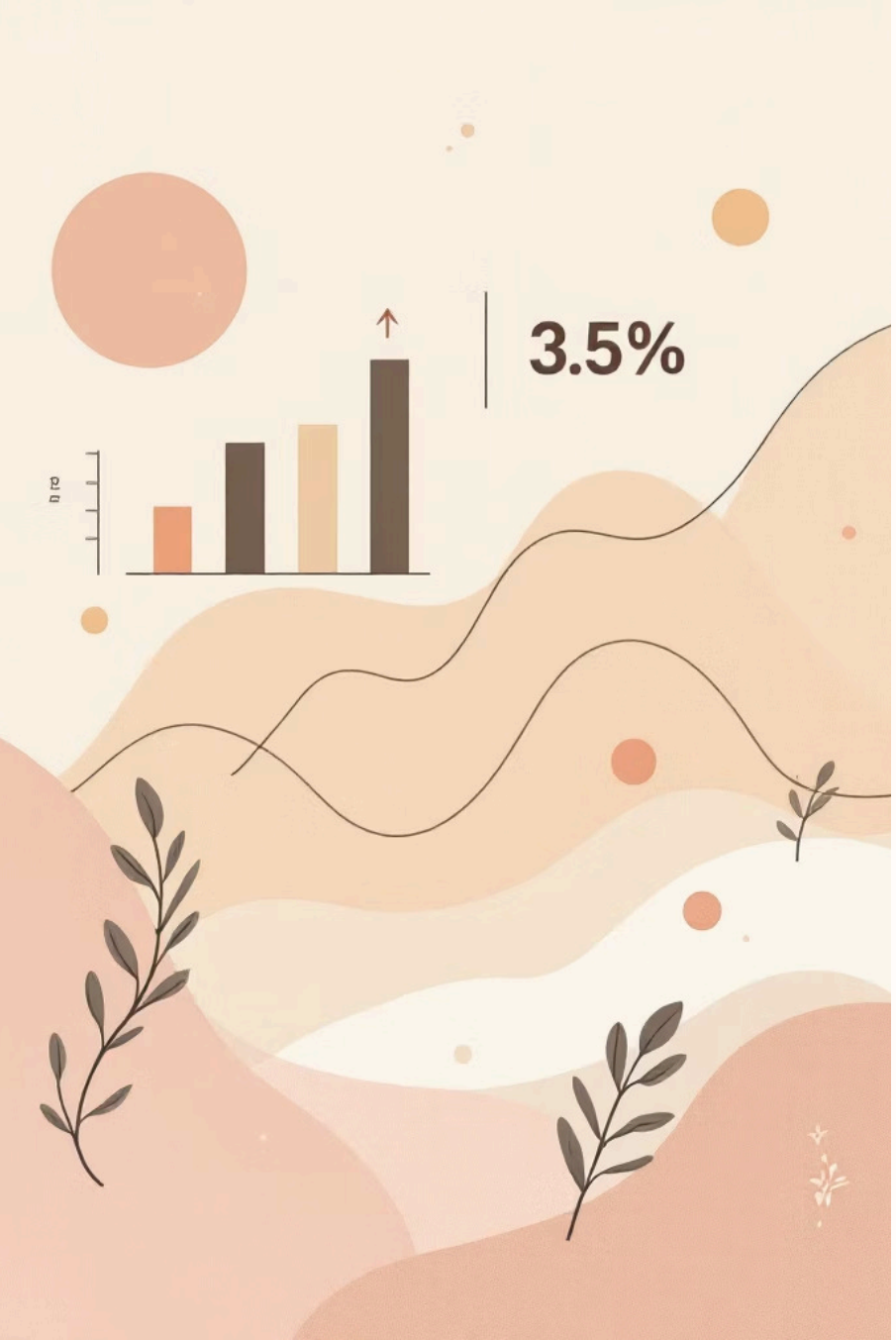
Data Cleaning

Handled missing values, standardized columns

Feature Engineering

Created age groups and frequency metrics





Revenue by Gender

Male Customers

Higher total revenue contribution

Female Customers

Strong purchasing power

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Discount Strategy Insights

High-Spending Discount Users

Identified customers using discounts while spending above average purchase amounts

Discount-Dependent Products

5 products with highest percentage of discounted purchases

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Product Performance

Top 5 Products by Rating

Highest average review ratings

Top 3 per Category

Most purchased products

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Shipping & Subscription Analysis

Shipping Comparison

Standard vs. Express shipping average purchase amounts

Subscribers vs. Non-Subscribers

Average spend and total revenue comparison

Repeat Buyers

Customers with >5 purchases more likely to subscribe

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Customer Segmentation

New

First-time customers

Returning

Multiple purchases

Loyal

Consistent buyers

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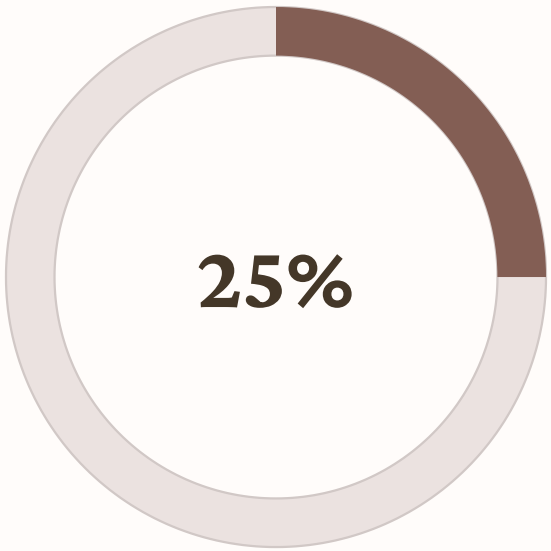
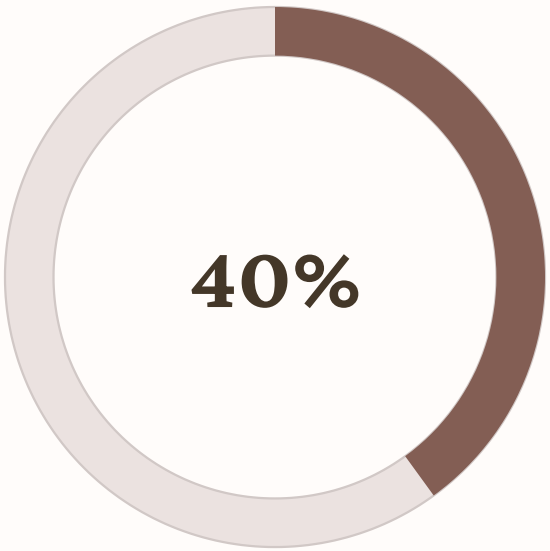
Revenue by Age Group



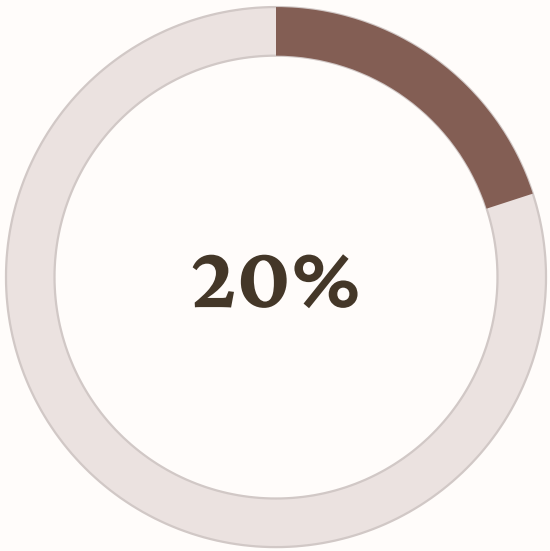
Young Adults

Age 18-30 segment



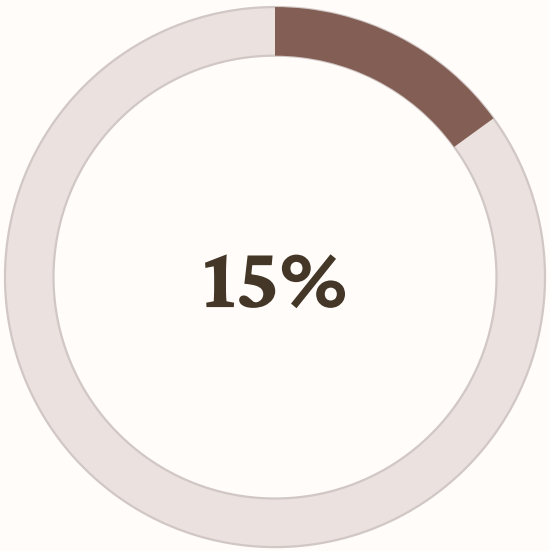
Prime Shoppers

Age 31-45 segment



Established

Age 46-60 segment



Senior

Age 60+ segment

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Strategic Recommendations

01

Boost Subscriptions

Promote exclusive benefits for subscribers

02

Customer Loyalty Programs

Reward repeat buyers to move into "Loyal" segment

03

Review Discount Policy

Balance sales boosts with margin control

04

Product Positioning

Highlight top-rated and best-selling products in campaigns

05

Targeted Marketing

Focus on high-revenue age groups and express-shipping users

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